Practical LaTeX Lab Guide – Part 1

Why LaTeX?

- Produces beautiful documents, especially for mathematics.
- Created by scientists, for scientists.
- Large and active community worldwide.
- Extendable with thousands of packages for papers, slides, bibliographies, tables, etc.

Installing LaTeX & Editor

- sudo apt update
- sudo apt install texlive-latex-extra -y
- sudo apt install texstudio -y

What each command does:

- sudo apt update → updates the list of available software (always run before installing).

- sudo apt install texlive-latex-extra -y
  → installs **LaTeX compiler** (TeX Live) + extra packages (needed for many documents).
  The -y means “yes to all prompts.”

- sudo apt install texstudio -y
  → installs **TeXstudio**, a graphical editor for LaTeX (you write .tex files here).

Steps after installation:

1. Verify installation

pdflatex --version

- If it prints version info (like pdfTeX 3.141592...), LaTeX is installed correctly.

2. Write your first document

- Open **TeXstudio** from Applications menu (or type in terminal):

  texstudio

- Inside TeXstudio, create a new file and type this (careful: no spaces in commands):

Write your first document: Create a file `hello.tex` with:

```
\documentclass{article}
\begin{document}
Hello, World! My first LaTeX document.
\end{document}
```

Save it as `hello.tex`.

- Output:-
- Hello, World! My first document.
- Paragraphs are separated by a blank line.
- Explanation:-
- • \documentclass{article} – Defines document type as an article.
- • \begin{document}. . . \end{document} – The main content of the document.
- • A blank line = new paragraph.

3. Compile the document

- In TeXstudio: press **F5** or click **Build & View** → it generates hello.pdf.

Or, from the terminal:

pdflatex hello.tex

- This creates a file hello.pdf in the same folder.

4. Open the output PDF

- xdg-open hello.pdf
- This will open the PDF in your Linux PDF viewer.

- **Optional tools**
- If later you need **bibliographies** or automatic builds:
- sudo apt install biber -y # bibliography support
- sudo apt install latexmk -y # auto compile

2. Document Class Options

Input:

```
\documentclass[11pt,a4paper,twocolumn]{article}
\begin{document}
This article uses 11pt font, A4 paper, and two columns.
\end{document}
```

Output (simulated):

This article uses 11pt font, A4 paper, and two columns.

This article uses 11pt font, A4 paper, and two columns.

- Explanations:- 11pt → Sets font size to 11pt. - a4paper → Paper size is A4. -
- twocolumn → Formats text in two columns.

3. Title, Author, Date

Input:

```
\documentclass{article}
\title{How to Structure a LaTeX Document}
\author{Leslie Lamport}
\date{December 1984}
\begin{document}
\maketitle
\end{document}
```

- Output:
- Hello from a report class! (styled as a report).
- Explanations:- \documentclass{report} → Defines the document as a report.
- - \begin{document}...\end{document} → Main body of the document.

How to Structure a LaTeX Document

Leslie Lamport

December 1984

- \title{...} → Defines document title. - \author{...} →Defines author name. - \date{...} → Adds date. - \maketitle → Prints the title block.

4. Sectioning Commands

Input:

```
\documentclass{article}
\begin{document}
\section{Main Section}
\subsection{Sub Section}
\subsubsection{Sub Sub Section}
\paragraph{Paragraph Title} Text here.
\subparagraph{Subparagraph Title} More text here.
\end{document}
```

Output (real):

1 Main Section

1.1 Sub Section

1.1.1 Sub Sub Section

Paragraph Title Text here.

Subparagraph Title More text here.

- Explanations:- \section{...} →Creates a main section. - \subsection{...}
- → Subsection under section. - \subsubsection{...} → Nested level deeper. -
- \paragraph{...} →Paragraph heading. - \subparagraph{...} →Sub-paragraph
- heading.

Lists

Input:

```
\begin{itemize}
  \item Biscuits
  \item Tea
\end{itemize}

\begin{enumerate}
  \item First item
  \item Second item
\end{enumerate}
```

Output:

- Biscuits
- Tea

- 1. First item
- 2. Second item

Explanation

Itemize

- itemize → creates **bullet points**.

Enumerate

- enumerate → creates a **numbered list**.

Items

- Each entry is written with \item.

5. Advanced Itemize

Input:

```
\documentclass{article}
\usepackage{amssymb} % Required in preamble
\begin{document}
\begin{itemize}
  \item Normal bullet
  \item[!] Exclamation point
  \item[$\blacksquare$] Black square
  \item[NOTE] Custom text
  \item[] Blank label
\end{itemize}
\end{document}
```

Output (real):

- Normal bullet

! Exclamation point

■ Black square

NOTE Custom text

Blank label

- Explanations:- \begin{itemize}.. .\end{itemize} → Creates bulleted list.
- - \item[custom] → Customizes the bullet. - \blacksquare → Special symbol
- (needs amssymb).

6. Advanced Enumerate

Input:

```
\documentclass{article}
\usepackage{amssymb} % Required in preamble
\begin{document}
\begin{enumerate}
  \item Normal numbered item
  \item[!] Custom exclamation
  \item[$\blacksquare$] Black square
  \item[NOTE] Custom text instead of number
  \item[] Blank label
\end{enumerate}
\end{document}
```

Output (real):

1. Normal numbered item

! Custom exclamation

■ Black square

NOTE Custom text instead of number

Blank label

- Explanations:- \begin{enumerate}.. .\end{enumerate} →Creates numbered list.

- \item[custom] → Replaces number with custom label.

7. Nested Lists

Input:

```
\documentclass{article}
\begin{document}
\begin{enumerate}
  \item First level
  \begin{itemize}
    \item Second level
    \begin{itemize}
      \item[Note:] Extra description
      \item[Wait!] Warning note
    \end{itemize}
  \end{itemize}
\end{enumerate}
\end{document}
```

Output (real):

1. First level
    - Second level

    Note: Extra description

    Wait! Warning note

- Explanations:- \begin{enumerate} → Numbered list.
- - \begin{itemize} →Bulleted list.
- - Nested lists allowed inside each other.

Input:

```
The rain in Spain falls \emph{mainly} on the plain.
He said, ``Hello, LaTeX!''
This costs \$10. The profit is 5\%.
```

Text Formatting and Special Characters

Output:

The rain in Spain falls *mainly* on the plain.
He said, “Hello, LaTeX!”
This costs $10. The profit is 5%.

Explanation

- \emph{mainly} → Produces **emphasis** (*mainly*).
- “ … ’’ → Produces **proper quotes** (“Hello, LaTeX!”).
- \$ → Prints the **$ symbol** (avoids math mode).
- \% → Prints the **% sign** (otherwise reserved in LaTeX).

8. Math Symbols

Input:

```
\documentclass{article}
\usepackage{amsmath,amssymb} % Required in preamble
\begin{document}
Inline: $a \pm b$, $x \times y$, $m \cdot n$, $p \div q$ \\
Relations: $x \ne y$, $n \le m$, $m \ge n$, $k \to \infty$ \\
Dots: $1,2,3,\ldots$ and $1+2+\cdots+n$ \\
Fraction: $\frac{a}{b}$
\end{document}
```

Output (real):

Inline: $a \pm b$, $x \times y$, $m \cdot n$, $p \div q$

Relations: $x \ne y$, $n \le m$, $m \ge n$, $k \to \infty$

Dots: $1, 2, 3, \ldots$ and $1 + 2 + \cdots + n$

Fraction: $\frac{a}{b}$

Explanations: - `\pm` → $\pm$ (plus/minus symbol).
- `\times` → $\times$ (multiplication cross).
- `\cdot` → $\cdot$ (multiplication dot).
- `\div` → $\div$ (division sign).
- `\ne` → $\neq$ (not equal).
- `\le` → $\leq$ (less than or equal).
- `\ge` → $\geq$ (greater than or equal).
- `\infty` → ∞ (infinity).
- `\ldots` → $\ldots$ (horizontal dots in text).
- `\cdots` → $\cdots$ (centered dots in math).
- `\frac{a}{b}` → $\frac{a}{b}$ (fraction).

Input:

```
% bad:
Let a and b be positive integers, and let c = a - b + 1.

% good:
Let $a$ and $b$ be positive integers, and let $c = a - b + 1$.
```

Inline Mathematics

Output (PDF Result):

- **Bad:** Let a and b be positive integers, and let c = a - b + 1.
- **Good:** Let *a* and *b* be positive integers, and let $c=a-b+1$.

Explanation

- **Inline math uses $ ... $.**
  Example: a → *a*
- **Math mode automatically adjusts style and spacing.**
  - Variables appear in *italic* (a, b, c).
  - Minus signs and equals signs look proper (− instead of -).

Good version (with $...$):

- Clear mathematical notation.
- Matches standard typesetting in books/papers.

Bad version (without $...$):

- Just plain text → looks unprofessional and incorrect.

Superscripts, Subscripts, and Greek Letters

Input:

```
$y = c_2 x^2 + c_1 x + c_0$
$F_{n} = F_{n-1} + F_{n-2}$
$\mu = A e^{Q/RT}$
$\Omega = \sum_{k=1}^{n} \omega_k$
```

Output:

$$y = c_2 x^2 + c_1 x + c_0$$
$$F_n = F_{n-1} + F_{n-2}$$
$$\mu = A e^{Q/RT}$$
$$\Omega = \sum_{k=1}^{n} \omega_k$$

Subscript (_)

- Example: c_2 → c_2
- Indexing works naturally for math variables.

Superscript (^)

- Example: x^2 → x^2
- Used for exponents and powers.

Grouping with {...}

- Example: F_{n-1} → F_{n-1}
- Keeps “n−1” together as a single subscript.
- Without {}, only the next symbol would be subscripted.

Greek letters (\mu, \Omega, \omega)

- \mu → **μ**
- \Omega → **Ω**
- \omega → **ω**

Summation (\sum)

- Example: \sum_{k=1}^{n} → **∑ (k=1 to n)**
- Limits (**k=1** to **n**) are placed correctly below/above the ∑.

Exponential (e^{...})

- Example: e^{Q/RT} → **e^(Q/RT)**
- Properly typesets scientific/engineering formulas.

Displayed Equations

Input:

```
\begin{equation}
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{equation}

\begin{equation*}
\Omega = \sum_{k=1}^{n} \omega_k
\end{equation*}
```

Output:

Numbered:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

Unnumbered:

$$\Omega = \sum_{k=1}^{n} \omega_k$$

Explanation

Equation environments

- equation → displays centered math **with numbering**.
- equation* → displays centered math **without numbering**.

Fractions

- \frac{a}{b} → **a⁄b**

Plus/Minus symbol

- \pm → **±**

Square root

- \sqrt{…} → **√(…)**

Figures

Input:

```
\begin{figure}[h]
\centering
\includegraphics[width=0.4\textwidth]{example-image}
\caption{An example figure}
\end{figure}
```

Output: (placeholder figure)



Figure 1: An example figure

Figure environment:-

- figure → wraps images inside a floating environment.

Centering:-

- \centering → centers the figure.

Including graphics:-

- \includegraphics[width=0.4\textwidth]{file} → scales image to **40% of text width**.

Caption:-

- \caption{...} → adds a caption below the figure.

Requirement:-

- Must add \usepackage{graphicx} in the preamble.

Error Handling

Input:

```
% Wrong:
\meph{wrong}

% Correct:
\emph{correct}
```

- **Output:** 5
- **Error (undefined command):** \meph{wrong}
- **Correct:** \emph{correct}

- **Advice:**
- Always fix the **first error** reported (later ones may be caused by it).
- Common errors:
  - Missing braces { }
  - Forgetting math mode $...$